



AQ6.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Choose the set of correct options.

☐

If v_1 and v_2 are in the kernel of a linear transformation $T: V \rightarrow W$, where V and W are two vector spaces, then $v_1 + v_2$ will also be in the kernel of T .

☐

Kernel of a linear transformation $T: V \rightarrow W$ is a vector subspace of V , where V and W are two vector spaces.

☐

If w_1 and w_2 are in the image of a linear transformation $T: V \rightarrow W$, where V and W are two vector spaces, then $w_1 + w_2$ will also be in the image of T .

☐

Image of a linear transformation $T: V \rightarrow W$ is a vector subspace of W , where V and W are two vector spaces.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and v_2 are in the kernel of a linear transformation $T: V \rightarrow W$, where V and W are two vector spaces, then $v_1 + v_2$ will also be in the kernel of T .

Kernel of a linear transformation $T: V \rightarrow W$ is a vector subspace of V , where V and W are two vector spaces.

If w_1 and w_2 are in the image of a linear transformation $T: V \rightarrow W$, where V and W are two vector spaces, then $w_1 + w_2$ will also be in the image of T .

Image of a linear transformation $T: V \rightarrow W$ is a vector subspace of W , where V and W are two vector spaces.

Consider the following linear transformation:

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$T(x, y, z) = (2x + 3z, 4y + z)$$

Answer questions 2,3,4 and 5, using the information given above.

1 point

Which of the following matrices corresponds to the given linear transformation T with respect to the standard ordered basis for $\mathbb{R}^3$ and the standard ordered basis for $\mathbb{R}^2$?

☐

$$\begin{bmatrix} 2 & 3 & 0 \\ 4 & 1 & 0 \end{bmatrix}$$

☐

$$\begin{bmatrix} 2 & 0 & 3 \\ 4 & 1 & 0 \end{bmatrix}$$

☐

$$\begin{bmatrix} 2 & 0 & 3 \\ 4 & 1 & 0 \end{bmatrix}$$



$$\begin{bmatrix} 24 \\ 31 \\ 00 \end{bmatrix} \begin{bmatrix} 230410 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 203 \\ 041 \end{bmatrix} \begin{bmatrix} 200431 \end{bmatrix}$$

1 point

Which of the following represents a basis of the kernel of T ?

☐

$$\left\{ \left(-\frac{3}{2}, 0, 1 \right), \left(0, -\frac{1}{4}, 1 \right) \right\} \left\{ (-23, 0, 1), (0, -41, 1) \right\}$$

☐

$$\left\{ \left(-\frac{3}{2}, -\frac{1}{4}, 1 \right) \right\} \left\{ (-23, -41, 1) \right\}$$

☐

$$\left\{ \left(-\frac{3}{2}, -\frac{1}{4}, 2 \right) \right\} \left\{ (-23, -41, 2) \right\}$$

☐

$$\left\{ (2, 0, 3), (0, 4, 1) \right\} \left\{ (2, 0, 3), (0, 4, 1) \right\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\left\{ \left(-\frac{3}{2}, -\frac{1}{4}, 1 \right) \right\} \left\{ (-23, -41, 1) \right\}$$

What will be the dimension of the subspace $\text{Im}(T)$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

Choose the correct option.

☐

T is an isomorphism.

☐

T is one to one but not onto.

☐

T is onto but not one to one.

☐

T is neither one to one, nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is onto but not one to one.

1 point

Choose the set of correct options.

☐

There exists a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range is contained in the circle $x^2 + y^2 = 1$.

☐

There exists a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range equals the line $x = y$.

☐

There exists a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range is contained in the set $S = \{(x, y) \mid x > 0\}$.

☐

There exists a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range equals the line $y = \pi x$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

There exists a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range equals the line $x = y$.

There exists a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range equals the line $y = \pi x$.

Level 2

1 point

Which option represents the kernel and image of the following linear transformation?

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x, 0)$$

☐

$$\ker(T) = \text{Span}\{(1, 0)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}$$

☐

$$\ker(T) = \text{Span}\{(1, 0)\}, \text{Im}(T) = \text{Span}\{(0, 1)\}$$

☐

$$\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}$$

☐

$$\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(0, 1)\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}$$

1 point

Which of the following linear transformations is an isomorphism between the vector spaces

$$V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in \mathbb{R}\} \subset \mathbb{R}^3$$

$$V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in \mathbb{R}\} \subset \mathbb{R}^3 \text{ and } W = \{(x, y, z) \mid x = y, \text{ and } x, y, z \in \mathbb{R}\} \subset \mathbb{R}^3$$

☐

$$T: V \rightarrow W, \text{ such that } T(x, y, z) = (y, y, 0)$$

☐

$$T: V \rightarrow W, \text{ such that } T(x, y, z) = (y, y, z)$$

☐

$$T: V \rightarrow W, \text{ such that } T(x, y, z) = (x, y, z)$$

☐

There does not exist any isomorphism between V and W .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T: V \rightarrow W$ $T: V \rightarrow W$, such that $T(x, y, z) = (y, y, z)$ $T(x, y, z) = (y, y, z)$

1 point

Which of the following linear transformations is an isomorphism between the vector spaces

$V = \{(x, y, z) \mid x = y - z = 0, \text{ and } x, y, z \in R\} \subset R^3$ $V =$

$\{(x, y, z) \mid x = y - z = 0, \text{ and } x, y, z \in R\} \subset R^3$ and

$W = \{(x, y, z) \mid x = y = 0, \text{ and } x, y, z \in R\} \subset R^3$ $W = \{(x, y, z) \mid x = y = 0, \text{ and } x, y, z \in R\}$
 $\subset R^3$?

☐

$T: V \rightarrow W$ $T: V \rightarrow W$, such that $T(x, y, z) = (x, 0, y)$ $T(x, y, z) = (x, 0, y)$.

☐

$T: V \rightarrow W$ $T: V \rightarrow W$, such that $T(x, y, z) = (0, y, z)$ $T(x, y, z) = (0, y, z)$.

☐

$T: V \rightarrow W$ $T: V \rightarrow W$, such that $T(x, y, z) = (0, 0, x)$ $T(x, y, z) = (0, 0, x)$.

☐

There does not exist any isomorphism between V and W .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T: V \rightarrow W$ $T: V \rightarrow W$, such that $T(x, y, z) = (x, 0, y)$ $T(x, y, z) = (x, 0, y)$.

1 point

Consider the linear transformation $T: R^3 \rightarrow R^3$ $T: R^3 \rightarrow R^3$ such that

$T(x, y, z) = (x - y + z, 2x + y - z, -x - 2y + 2z)$ $T(x, y, z) = (x - y + z, 2x + y - z, -x - 2y + 2z)$.

Which of the following options are true about T ?

☐

Range of T is $\{(x, y, z) \in R^3 \mid x = y + z\}$ $\{(x, y, z) \in R^3 \mid x = y + z\}$.

☐

Range of T is $\{(x, y, z) \in R^3 \mid x = 0, y + z = 0\}$ $\{(x, y, z) \in R^3 \mid x = 0, y + z = 0\}$.

☐

Rank of T is 2

☐

Rank of T is 1

No, the answer is incorrect.

Score: 0

Accepted Answers:

Range of T is $\{(x, y, z) \in R^3 \mid x = y + z\}$ $\{(x, y, z) \in R^3 \mid x = y + z\}$.

Rank of T is 2

[Check Answers](#)

Your score is: 0/10